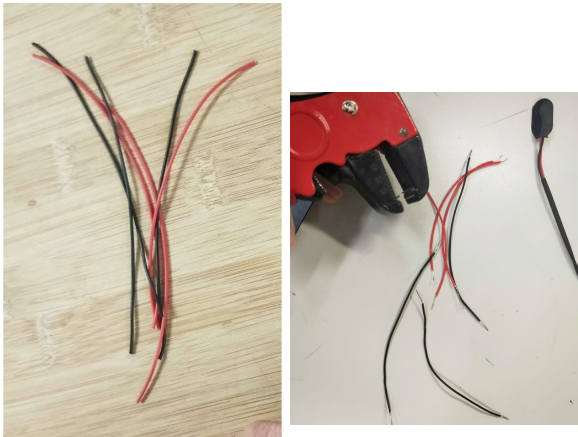


Heads or Tails Manufacturing Process

By Nicolas

In this manufacturing process, I will explain what I have done to create a Heads or tails game. The process was divided into 3 sections; creating the circuit, creating the housing and creating the user interface + assembling the product.

Photo	Step
Creating the circuit	
	Step 1 - Assembling the circuit board In this step, I added all the circuit components to the circuit board. I made sure for all of the resistors that the golden ring would be pointing to the top of the board to make sure they're pointing in the same direction. Remember to bend the wires after attaching them to keep the components in place. Materials required: - Circuit board - Resistors - Capacitors - LEDs and transistors Things to keep in mind/safety: - Keep all the components organised so you don't lose any components
	Step 2 - Cutting and stripping wires In this step, we stripped 3 red wires, 3 black wires and 2 green wires (not in photo) and cut them down to 12cm each. We also stripped the ends of a pair of wires connecting to the battery clip. Materials required: - 3 black wires - 3 red wires - 2 green wires - Wire stripper - Battery clip with wires Things to keep in mind/safety: - Don't strip the wire too long. You can always strip again if needed - Make sure all the wires are about 12 cm so they line up when you solder them



Step 3 - Soldering

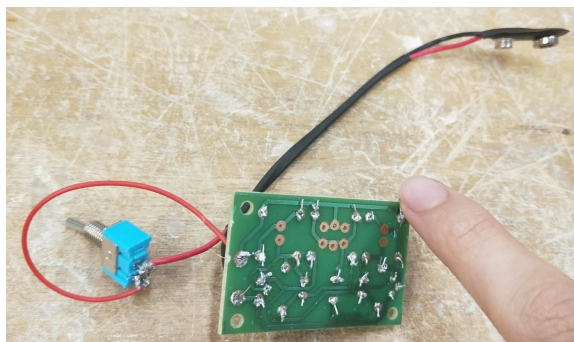
In this step, I soldered the components from step 1 to the circuit board. I cut the wires holding the components in place after soldering.

Materials required:

- Circuit board and components from step 1
- Wirecutter
- Solder
- Soldering iron
- Soldering station

Things to keep in mind/safety:

- Don't burn yourself
- Avoid inhaling the smoke
- Wear goggles
- Use the vice to clamp the edge of the circuit board for more efficiency and precision
- You shouldn't be able to see any more copper after soldering



Step 4 - Attaching the switch and battery clip to the circuit

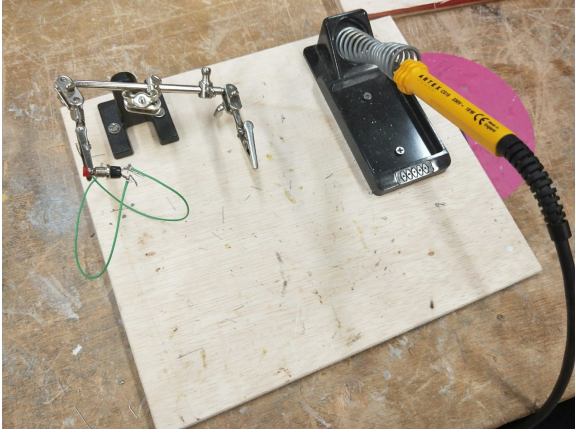
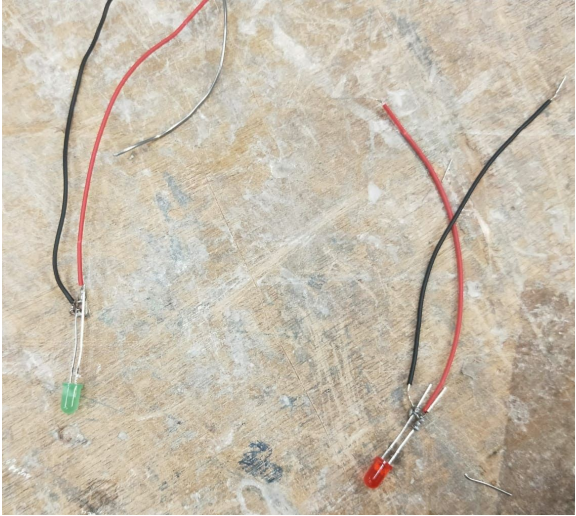
For this step, I soldered one red wire connecting a single pole double throw switch (SPST) and the +9V hole on the right side of the circuit board together and connecting another red wire coming from the battery clip to the switch. I then soldered the remaining black wire connecting the battery clip and the G hole right below the 9V+ hole.

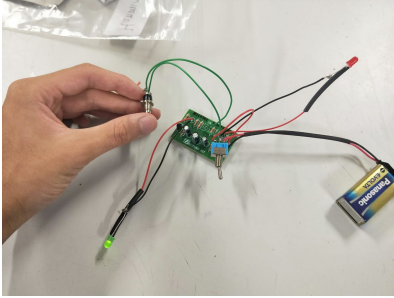
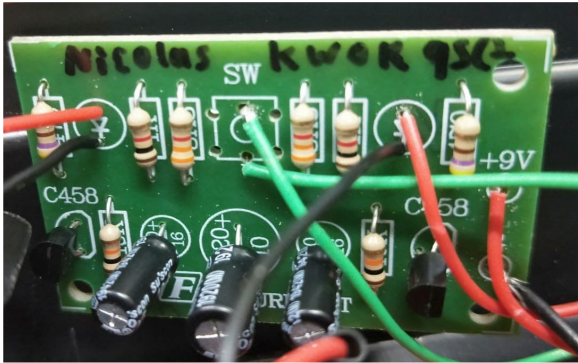
Materials required:

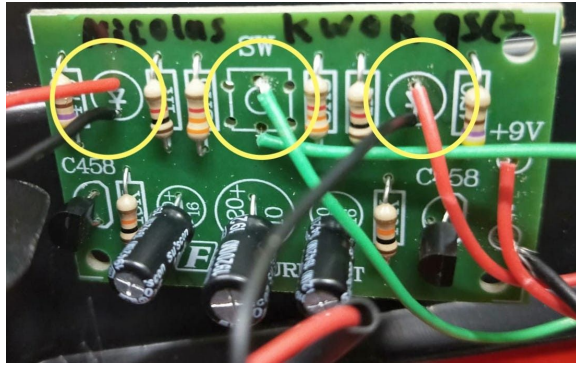
- Circuit board and components from step 3
- Solder
- Soldering iron
- Soldering station
- Red wire
- Battery clip

Things to keep in mind/safety:

- Don't burn yourself
- Avoid inhaling the smoke
- Wear goggles
- Use the vice to clamp the circuit

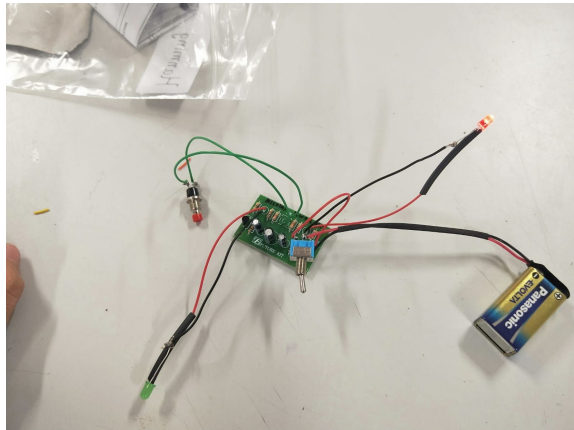
	board for more efficiency and precision - You shouldn't be able to see any more copper after soldering - Make sure you solder the switch firmly, to avoid the risk of having the wire detach
	Step 5 - Soldering green wires to the button In this step, I soldered 2 green wires from step 2 to a push to make (PTM) switch or button. Materials required: - Button - 2 green wires - Solder - Soldering iron - Soldering station Things to keep in mind/safety: - Don't burn yourself - Avoid inhaling the smoke - Wear goggles - Use the vice to clamp the button for more efficiency and precision - Make sure you solder the button firmly, to avoid the risk of having the wire detach
	Step 6 - Soldering the LED For this step, I soldered the red wires to the positive side (longer) and the black wires to the negative side (shorter) of the LED. Repeat this step twice, 1 time for the red LED and 1 time for the green LED. Materials required: - Green LED - Red LED - 2 red wires - 2 black wires - Solder - Soldering iron - Soldering station Things to keep in mind/safety: - Don't burn yourself - Avoid inhaling the smoke - Wear goggles - Use the vice to clamp the LED for more efficiency and precision

	- Make sure you solder the LED firmly, to avoid the risk of having the wire detach
 	Step 7 - Put a heat shrink on a wire connecting to the LED For this step, I cut a small tube of heat shrink with scissors and put it on a wire connecting to the LED. This is to avoid a short circuit from touching wires. After the heat shrink is in place, apply heat for about 10 seconds on all sides. Materials required: - Heat shrink - Scissors - Heat gun - LED with wires connected Things to keep in mind/safety: - Don't point the gun towards your hand - Don't apply heat for too long - Don't put your hand too close to the heat source
 	Step 8 - Connect the LEDs and the button to the circuit board In this step, I connected the button to 2 holes below the box SW (look at green wires). I also soldered the LED wires in the 2 circles between the resistors. I soldered the red wire to the end with the triangle and the black wire to the end with the line. Materials required: - Circuit board - Green LED with wires - Red LED with wires - Button with wires - Solder - Soldering iron - Soldering station Things to keep in mind/safety: - Don't burn yourself - Avoid inhaling the smoke - Wear goggles - Use the vice to clamp the circuit board for more efficiency and precision - Make sure you solder the



components firmly, to avoid the risk of having the wire detach

[Link to video](#)



Step 9 - Test the circuit

Make sure the circuit works before proceeding, attach a **9-volt** battery to the battery clip and turn on the switch. Use the button to start each run. If the circuit works like in the video, then keep going and continue but if the circuit doesn't then you may have to polish up your soldering and troubleshoot

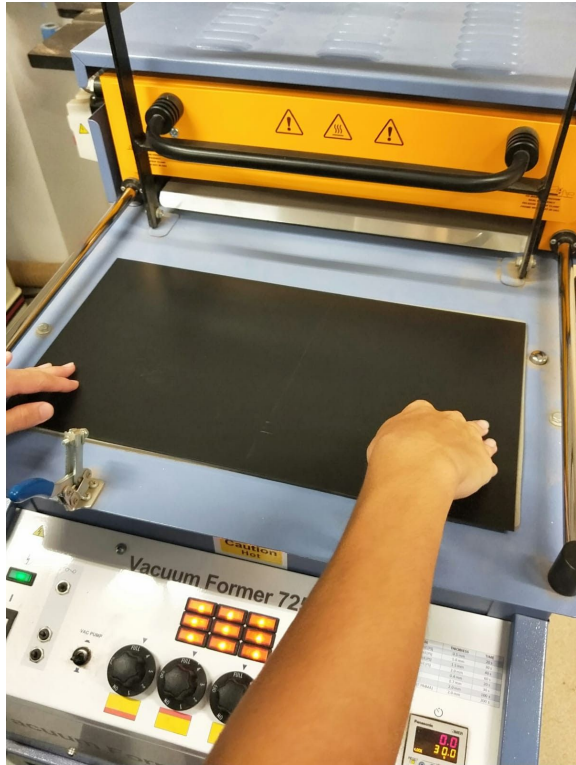
Materials required:

- Completed circuit board
- 9-volt battery

Things to keep in mind/safety:

- Don't use more than 9 volts

Creating the housing



Step 10 - Making the housing

In this step, we used a high-density polystyrene sheet in a vacuum former. First, peel off the protective layer on the sheet (if there is one) and place it on the vacuum former. Then clamp it down to keep it in place

Materials required:

- Vacuum former
- High-density polystyrene sheet

Things to keep in mind/safety:

- Make sure the protective layer is peeled off if there is one
- Turn on the machine 10 - 15 minutes before you heat it
- Make sure the sheet of polystyrene is clamped



Step 11 - Heating up the polystyrene

In this step, I pulled the heater over the sheet of polystyrene to heat it up for 30 seconds.

Materials required:

- Vacuum former
- High-density polystyrene sheet

Things to keep in mind/safety:

- Heat it for the correct amount of time, not too long not too short
- Turn on the machine 10 - 15 minutes before you heat it



Step 12 - Moulding the sheet of polystyrene

For this following step, I pushed back the heater and pushed the handle up to mould the polystyrene sheet (photo on the right), I then removed the air from underneath the sheet of polystyrene (photo on the left).

Materials required:

- Vacuum former
- High-density polystyrene sheet

Things to keep in mind/safety:

- Beware that the sheet of polystyrene may be hot after removal
- Don't be too slow



Step 13 - Cutting and sanding the housing

For this step, I used a craft knife to roughly cut out the shape of the housing. I then used a sanding board to sand down the unwanted edges.

Materials required:

- Sanding board
- Craft knife
- High-density polystyrene sheet (molded)

Things to keep in mind/safety:

- Cut away from your body with the craft knife
- Try to even out the sanding per side so it's even.



Photo from Satvik



My second layer of housing is red because I sanded down the first one too much

Step 14 - Drilling the hole for the screw

For this step, I started off by measuring the dimensions of the housing (both layers) and marked the spot where I wanted to drill a hole. The mark should be the same position for both layers and should be centered in the 2nd layer. Then, I used a hole puncher to puncture a hole through the first housing and drilled the second layer using a wooden dowel and a clamp to keep the piece of housing in place. I then used an M4 hex nut and an M4 screw to test out the hole and to make sure that the hole size is correct.

Materials required:

- Wooden dowel
- Clamp
- Drill
- Housing
- 2 M4 hex nuts
- 2 M4 screws
- Hole puncher

Things to keep in mind/safety:

- Be careful with the drill
- Don't rush to puncture the hole, you only have one chance
- Make sure the wooden dowel is clamped tightly

Creating the user interface + assembling the product



Step 15 - Making the user interface

In this step, I created a user interface made out of acrylic using a laser cutter. I sent a template from the computer to laser cut. After finishing the laser cutting, I used a black marker to colour over the words. Then I peeled off the top layer of plastic to keep the words darkened.

Materials required:

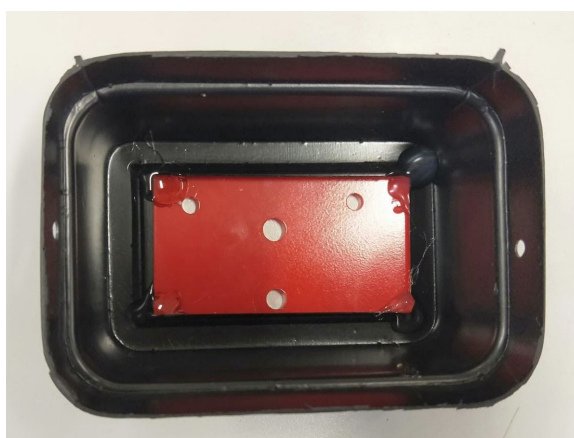
- Laser cutter
- Acrylic
- Template for user interface
- Marker

Things to keep in mind/safety:



(only top part)

- Position the acrylic correctly in the laser cutter machine
- Use a different coloured marker to colour over the text



Step 16 - Glueing the user interface to the case

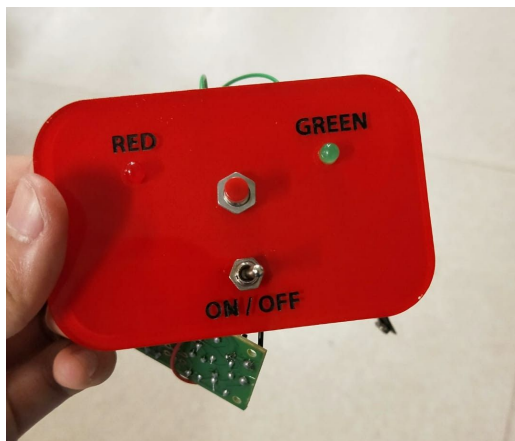
In this step, I glued the user interface to the case using a hot glue gun.

Materials required:

- Housing
- User interface
- Hot glue gun

Things to keep in mind/safety:

- Make sure the user interface is in place and aligned with the housing
- Make sure the glue doesn't enter the hole
- Make sure the glue covers both the housing and the user interface
- Don't burn yourself
- Try to avoid leaving strands of glue



Step 17 - Finishing off

For this final step, I put the circuit with the battery connected in the housing with correct components sticking out from the correct holes. For the switch and the button, I used one hex nut each to keep it in place. Test the circuit to see if it works. If it doesn't, troubleshoot by looking at the soldering and see if there are any faults. Then secure the circuit in the housing with a hex nut and screw.

Materials required:

- Housing + user interface
- Circuit
- 2 hex nut's different sizes about 0.5cm each side



- Battery

Things to keep in mind/safety:

- Try not to break the wires in the circuit while putting it in